Block practice · Data types, variables, arrays and subprograms

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Pick the right data type for a value

Variables, constants, arrays and records

Functions, procedures, parameters and libraries

Local and global variables

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Practice workbook · Pick the right data type for a value

Recognising which primitive data type suits a value or variable, giving example values for a type, and stating the values a Boolean can hold.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T2-01

1. Question 1 · part e · 2 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part e only.

1 Programmers write programs to solve problems.

(a) Programmers use many different techniques and tools to solve problems.

Identify what is meant by the term **abstraction**.

(1)

- ☒ **A** Breaking a problem down into smaller more manageable parts
- ☒ **B** Deciding if a program is efficient in terms of execution time
- ☒ **C** Drawing flowcharts with symbols used in the computer industry
- ☒ **D** Removing unnecessary detail to highlight the important points

(b) Open **Q01b** in the code editor.

A user enters a name and an age when the program executes.

The program should display a welcome message when the user is less than 30 years of age.

Amend the code to complete the program.

Save your amended code as **Q01bFINISHED** with the correct file extension for the programming language.

(4)

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1. Question 1 · part e · 2 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part e only.

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(c) Open **Q01c** in the code editor.

Use the code to answer these questions.

(i) Give the text of a comment used in the code.

(1)

(ii) Give the text of a line that creates and initialises a variable.

(1)

(iii) Give the keyword that starts the selection used in the code.

(1)

(iv) Give the logical operator used in the code.

(1)

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(d) Arrays and records are data structures.

Identify the data that **must be** stored in a record rather than an array.

(1)

- ☒ **A** "King Lear", "Macbeth", "Romeo and Juliet", "The Tempest"
- ☒ **B** "King Lear", 1983, 78.32, 99
- ☒ **C** 1978, 1985, 1990, 2001
- ☒ **D** 13.99, 12.75, 10.58, 11.43

(e) Complete the table to show the data type for **each** item.

(2)

Item	Data type
45.82	
True	

(Total for Question 1 = 12 marks)

1. Question 1 · part e · 2 marks · Working and answer

Working and answer

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2. Question 1 · part a · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part a only.

1 Computer programs use several different programming constructs.

(a) Identify the **most appropriate** data type for the value 4.5

(1)

- ☒ **A** Boolean
- ☒ **B** Character
- ☒ **C** Real
- ☒ **D** String

(b) Give a definition of a logic error.

(1)

(c) Open **Q01c** in the code editor.

The program should print out a counter and the counter with 7 added to it.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

(d) Give **one** reason why computer programmers should use comments in their programs.

(1)

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2. Question 1 · part a · 1 mark · Working and answer

Working and answer

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3. Question 2 · part b.iii · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part b.iii only.

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2 Anna has a 5-year-old daughter, Beatrice, and a 14-year-old son, Graham.

(a) Anna wants to program a guessing game for Beatrice to play on the computer.

The program will generate a random number between 1 and 10.

Beatrice has to guess the number.

This pseudocode contains the logic required to create the game.

```
1   #   Initialise variables
2       SET counter TO 1
3       SET answer TO RANDOM(9) + 1     # i.e. a random integer 1 to 10 inclusive
4       SET guess TO 0
5
6   #   Print prompt and take guess from user
7       SEND "Enter a number from 1 to 10: " TO DISPLAY
8       RECEIVE guess FROM (INTEGER) KEYBOARD
9
10  #   Create while loop to check guess
11      WHILE guess <> answer DO
12          SET counter TO counter + 1
13          IF guess > answer THEN
14              SEND (guess & " was too high. Try again.") TO DISPLAY
15          ELSE
16              SEND (guess & " was too low. Try again.") TO DISPLAY
17          END IF
18          SEND "Guess again: " TO DISPLAY
19          RECEIVE guess FROM (INTEGER) KEYBOARD
20      END WHILE
21
22  #   Report the correct answer to the user and display the number of guesses
23      SEND ("You guessed " & guess & " in " & counter & "guesses.") TO DISPLAY
```

Write a program to implement the logic in the pseudocode.

Open **Q02a** in the code editor.

You **must** use the structure give in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

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3. Question 2 · part b.iii · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part b.iii only.

(b) Anna has written a program to help Graham with calculations for his science lessons.

Open **Q02b** in the code editor.

Use the code to answer these questions.

(i) State the value of **distance** when the program finishes.

(1)

(ii) Give **one** reason why there are brackets around **newSpeed + initialSpeed**

(1)

(iii) State the data type for the variable **distance**.

Justify your answer.

(2)

Data type

Justification

(iv) State the value of **distance** when **acceleration** is 2.75

(1)

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3. Question 2 · part b.iii · 2 marks · Working and answer

Working and answer

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4. Question 1 · part b · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part b only.

1 Computer programs make use of many programming constructs.

(a) Identify which **one** of these is **not** an arithmetic operator.

(1)

- ☒ **A** add
- ☒ **B** divide
- ☒ **C** equals
- ☒ **D** modulus

(b) State the values used by the Boolean data type.

(1)

.....

.....

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4. Question 1 · part b · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part b only.

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(c) Open **Q01c** in the code editor.

The program should multiply two integers.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

(d) State what is meant by the term 'variable'.

(1)

(e) Ayesha has written a program that calculates the sum of a set of numbers.

Open **Q01e** in the code editor.

Answer these questions about the code.

(i) Identify the line number where the code includes a logic operator.

(1)

(ii) Identify the line number where a subprogram definition starts.

(1)

(iii) Identify the name of a numeric variable.

(1)

(iv) Identify the name of a string variable.

(1)

(Total for Question 1 = 10 marks)

4. Question 1 · part b · 1 mark · Working and answer

Working and answer

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5. Question 1 · part c · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.

1 Programmers write code to solve problems.

(a) Identify the description of a variable in a computer program.

(1)

- ☒ **A** A value that cannot be used more than once
- ☒ **B** A value that must be input
- ☒ **C** A value that is always used in a calculation
- ☒ **D** A value that can change

(b) Identify the technique that improves the readability of code.

(1)

- ☒ **A** Using indents on every line
- ☒ **B** Using descriptive names for variables
- ☒ **C** Using the correct operators
- ☒ **D** Using suitable data structures

(c) Complete the table by giving an example value for **each** data type.

The first row has been completed for you.

(2)

Data type	Example value
integer	12
char	
real	

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5. Question 1 · part c · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.

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(d) Describe **one** difference between the data used in boundary testing and the data used in erroneous testing.

(2)

(e) A program should output the value 2

However, there is an error in the code and the actual output is 4

Name this type of error.

(1)

(f) Open **Q01f** in the code editor.

The program should allow the input of the length of the side of a square and output the area of the square.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01fFINISHED** with the correct file extension for the programming language.

(3)

(g) A program is needed that will accept the input of a letter and compare it with a stored letter.

It will check whether the letter comes earlier in the alphabet, later in the alphabet or is the same letter as the stored letter.

It will output the letter and the stored letter with an appropriate message.

Open **Q01g** in the code editor.

Amend the code to complete the if statement used to produce the output.

You must use the structure given in **Q01g** to complete the program.

Do not add any further functionality.

Save your code as **Q01gFINISHED** with the correct file extension for the programming language.

(4)

(Total for Question 1 = 14 marks)

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5. Question 1 · part c · 2 marks · Working and answer

Working and answer

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6. Question 2 · part b.i · 2 marks

2025 JUN · Paper 02

Further practice · Attempt part b.i only.

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2 Programs can process numerical data.

(a) Identify the method used to check input data matches set rules.

(1)

- ☒ **A** Abstraction
- ☒ **B** Encryption
- ☒ **C** Iteration
- ☒ **D** Validation

(b) **Figure 2** shows a section from an employee payslip produced by a computer program.

Pay Rate	Hours		Total Pay
£ 18.75	19		£ 356.25
	Tax	20%	£ 71.25
		Income	£ 285.00

Figure 2

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(i) Complete the table to give **one** data type shown in **Figure 2** and give an example value.

(2)

Data type	
Example	

(ii) Complete the table to give an input, a process and an output used in the program to produce the payslip.

(3)

Input	
Process	
Output	

P 7 8 6 3 9 A 0 5 2 0

6. Question 2 · part b.i · 2 marks · Working and answer

Working and answer

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Practice workbook · Variables, constants, arrays and records

Definitions, purposes and benefits of variables and constants, and how arrays (1D/2D) and records differ from single variables and from each other.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T2-02

1. Question 1 · part c.ii · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part c.ii only.

1 Programmers use different programming constructs to create working code.

(a) Identify the keyword used to create a selection statement.

(1)

- ☒ **A** for
- ☒ **B** if
- ☒ **C** new
- ☒ **D** while

(b) Computer code can contain errors.

(i) Complete the table by adding a tick (✓) to match each error description to a type of error.

(3)

Description	Syntax error	Logic error	Runtime error
The program does not produce the expected output.			
The program does not translate.			
The program crashes during execution.			

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1. Question 1 · part c.ii · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part c.ii only.

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(ii) A program counts the number of green and red balls stored in an array.

The program outputs the number of green balls, the number of red balls and the total number of balls.

Figure 1 shows the expected output from the program.

```
Green:  25 balls
Red:  15 balls
Total:  40 balls
```

Figure 1

Open **Q01bii** in the code editor.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01biiFINISHED** with the correct file extension for the programming language.

(3)

(c) Programs use constants, variables and data structures.

(i) Give **one** reason for using a constant.

(1)

(ii) Describe **one** difference between a variable and an array.

(2)

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1. Question 1 · part c.ii · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part c.ii only.

(d) A program is used to calculate wages for sales staff. Staff can earn 5% commission based on their age and sales.

Open **Q01d** in the code editor.

Use the code to answer these questions.

(i) Identify the name of a user-written subprogram.

(1)

(ii) Identify a logical operator used in this program.

(1)

(iii) Identify the name of a variable that holds a real number.

(1)

(iv) Identify the name of a parameter.

(1)

(Total for Question 1 = 14 marks)

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Working and answer

2. Question 1 · part d · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part d only.

1 Computer programs make use of many programming constructs.

(a) Identify which **one** of these is **not** an arithmetic operator.

(1)

- ☒ **A** add
- ☒ **B** divide
- ☒ **C** equals
- ☒ **D** modulus

(b) State the values used by the Boolean data type.

(1)

.....

.....

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2

P 6 1 8 8 5 A 0 2 1 6

2. Question 1 · part d · 1 mark

2020 NOV · Paper 02

Further practice · Attempt part d only.

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(c) Open **Q01c** in the code editor.

The program should multiply two integers.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

(d) State what is meant by the term 'variable'.

(1)

.....

.....

(e) Ayesha has written a program that calculates the sum of a set of numbers.

Open **Q01e** in the code editor.

Answer these questions about the code.

(i) Identify the line number where the code includes a logic operator.

(1)

.....

(ii) Identify the line number where a subprogram definition starts.

(1)

.....

(iii) Identify the name of a numeric variable.

(1)

.....

.....

(iv) Identify the name of a string variable.

(1)

.....

.....

(Total for Question 1 = 10 marks)

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2. Question 1 · part d · 1 mark · Working and answer

Working and answer

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3. Question 1 · part b · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

1 Computer programs make use of many programming constructs.

(a) Identify which **one** of these is a logic operator.

(1)

- ☒ **A** modulus
- ☒ **B** OR
- ☒ **C** >=
- ☒ **D** =

(b) Describe **one** difference between a one-dimensional array and a two-dimensional array.

(2)

(c) Open **Q01c** in the code editor.

The program should calculate how many £20 are in £113 and how much is left over.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

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2

P 6 6 4 9 0 A 0 2 1 6

3. Question 1 · part b · 2 marks · Working and answer

Working and answer

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4. Question 1 · part a · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part a only.

1 Programmers write code to solve problems.

(a) Identify the description of a variable in a computer program.

(1)

- ☒ **A** A value that cannot be used more than once
- ☒ **B** A value that must be input
- ☒ **C** A value that is always used in a calculation
- ☒ **D** A value that can change

(b) Identify the technique that improves the readability of code.

(1)

- ☒ **A** Using indents on every line
- ☒ **B** Using descriptive names for variables
- ☒ **C** Using the correct operators
- ☒ **D** Using suitable data structures

(c) Complete the table by giving an example value for **each** data type.

The first row has been completed for you.

(2)

Data type	Example value
integer	12
char	
real	

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4. Question 1 · part a · 1 mark · Working and answer

Working and answer

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5. Question 1 · part c.i · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part c.i only.

1 Programs are used to handle financial transactions.

(a) Monthly account statements are created by a program.

Figure 1 shows a statement.

Date	Description	Debit (£)	Credit (£)	Balance (£)
01 Jan	Opening balance			128.35
04 Jan	Music store	10.23		118.12
07 Jan	Salary		1,515.28	1,633.40
11 Jan	Electricity company	50.00		1,583.40
19 Jan	Uniform shop	12.56		1,570.84
27 Jan	Refund		25.00	1,595.84
29 Jan	Rent	750.00		845.84
31 Jan	Interest		0.25	846.09

Figure 1

Complete the table to identify an input, an output and a process used by the program to generate the **furthest right column**.

(3)

Input	
Processing	
Output	

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5. Question 1 · part c.i · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part c.i only.

DO NOT WRITE IN THIS AREA

(b) A tax rate is applied to a gross value to give a net value.

Open **Q01b** in the code editor.

Use the code to answer these questions.

(i) Give the contents of a comment.

(1)

(ii) Identify a logical operator used in this program.

(1)

(iii) Give the name of a global variable.

(1)

(iv) Give the name of a constant.

(1)

(v) Give the name of a parameter.

(1)

(c) Variables and constants are used in program code.

(i) State the purpose of a constant.

(1)

(ii) Give **one** benefit of using a constant.

(1)

(Total for Question 1 = 10 marks)

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5. Question 1 · part c.i · 1 mark · Working and answer

Working and answer

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6. Question 1 · part c.ii · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part c.ii only.

1 Programs are used to handle financial transactions.

(a) Monthly account statements are created by a program.

Figure 1 shows a statement.

Date	Description	Debit (£)	Credit (£)	Balance (£)
01 Jan	Opening balance			128.35
04 Jan	Music store	10.23		118.12
07 Jan	Salary		1,515.28	1,633.40
11 Jan	Electricity company	50.00		1,583.40
19 Jan	Uniform shop	12.56		1,570.84
27 Jan	Refund		25.00	1,595.84
29 Jan	Rent	750.00		845.84
31 Jan	Interest		0.25	846.09

Figure 1

Complete the table to identify an input, an output and a process used by the program to generate the **furthest right column**.

(3)

Input	
Processing	
Output	

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6. Question 1 · part c.ii · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part c.ii only.

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DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) A tax rate is applied to a gross value to give a net value.

Open **Q01b** in the code editor.

Use the code to answer these questions.

(i) Give the contents of a comment.

(1)

(ii) Identify a logical operator used in this program.

(1)

(iii) Give the name of a global variable.

(1)

(iv) Give the name of a constant.

(1)

(v) Give the name of a parameter.

(1)

(c) Variables and constants are used in program code.

(i) State the purpose of a constant.

(1)

(ii) Give **one** benefit of using a constant.

(1)

(Total for Question 1 = 10 marks)

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6. Question 1 · part c.ii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 1 · part d · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part d only.

1 Programmers write programs to solve problems.

(a) Programmers use many different techniques and tools to solve problems.

Identify what is meant by the term **abstraction**.

(1)

- ☒ **A** Breaking a problem down into smaller more manageable parts
- ☒ **B** Deciding if a program is efficient in terms of execution time
- ☒ **C** Drawing flowcharts with symbols used in the computer industry
- ☒ **D** Removing unnecessary detail to highlight the important points

(b) Open **Q01b** in the code editor.

A user enters a name and an age when the program executes.

The program should display a welcome message when the user is less than 30 years of age.

Amend the code to complete the program.

Save your amended code as **Q01bFINISHED** with the correct file extension for the programming language.

(4)

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DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

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7. Question 1 · part d · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part d only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

(c) Open **Q01c** in the code editor.

Use the code to answer these questions.

(i) Give the text of a comment used in the code.

(1)

(ii) Give the text of a line that creates and initialises a variable.

(1)

(iii) Give the keyword that starts the selection used in the code.

(1)

(iv) Give the logical operator used in the code.

(1)

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(d) Arrays and records are data structures.

Identify the data that **must be** stored in a record rather than an array.

(1)

- ☒ **A** "King Lear", "Macbeth", "Romeo and Juliet", "The Tempest"
- ☒ **B** "King Lear", 1983, 78.32, 99
- ☒ **C** 1978, 1985, 1990, 2001
- ☒ **D** 13.99, 12.75, 10.58, 11.43

(e) Complete the table to show the data type for **each** item.

(2)

Item	Data type
45.82	
True	

(Total for Question 1 = 12 marks)

7. Question 1 · part d · 1 mark · Working and answer

Working and answer

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8. Question 2 · part a.ii · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part a.ii only.

2 Programmers design, correct and test programs.

(a) Programmers design programs.

(i) Identify the term for a step-by-step description to complete a task.

(1)

- ☒ **A** Algorithm
- ☒ **B** Computational thinking
- ☒ **C** Decomposition
- ☒ **D** Pseudocode

(ii) A constant is a memory location whose value does not change during program execution.

Give the name for a memory location whose value can change during program execution.

(1)

.....

.....

(b) Programs can have runtime errors.

(i) State what is meant by the term **runtime error**.

(1)

.....

.....

(ii) Give **one** example of a runtime error.

(1)

.....

.....

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8. Question 2 · part a.ii · 1 mark · Working and answer

Working and answer

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9. Question 1 · part c.i · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part c.i only.

1 Programmers use different programming constructs to create working code.

(a) Identify the keyword used to create a selection statement.

(1)

- ☒ **A** for
- ☒ **B** if
- ☒ **C** new
- ☒ **D** while

(b) Computer code can contain errors.

(i) Complete the table by adding a tick (✓) to match each error description to a type of error.

(3)

Description	Syntax error	Logic error	Runtime error
The program does not produce the expected output.			
The program does not translate.			
The program crashes during execution.			

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9. Question 1 · part c.i · 1 mark

2025 JUN · Paper 02

Further practice · Attempt part c.i only.

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(ii) A program counts the number of green and red balls stored in an array.

The program outputs the number of green balls, the number of red balls and the total number of balls.

Figure 1 shows the expected output from the program.

```
Green:  25 balls
Red:  15 balls
Total:  40 balls
```

Figure 1

Open **Q01bii** in the code editor.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01biiFINISHED** with the correct file extension for the programming language.

(3)

(c) Programs use constants, variables and data structures.

(i) Give **one** reason for using a constant.

(1)

(ii) Describe **one** difference between a variable and an array.

(2)

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9. Question 1 · part c.i · 1 mark · Working and answer

Working and answer

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Practice workbook · Functions, procedures, parameters and libraries

Why subprograms and libraries are used, what a built-in subprogram is, how functions and procedures are alike and differ, what a function returns, and why parameters are used.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T2-04

1. Question 5 · part c.ii · 2 marks

2020 NOV · Paper 02

Worked-example retry · Attempt part c.ii only.

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5 Cerys Jones is the manager of a shop that sells electrical goods.

She is creating a stock control program.

(a) Each item of stock will have a ten-character code.

Each code consists of three parts.

Part	Description
Characters 1 to 3	Uppercase letters
Characters 4 to 9	Numbers between 1 and 9 inclusive
Character 10	'e' or 'o' indicating whether the sum of the numbers is even or odd

WAS123456o, FDG232324e and VAC129222e are valid codes.

All codes need to be validated.

Complete the table to show **two** additional validation tests.

For each test, give **one** example of erroneous data. The example that you give should fail **only** that test.

(4)

Validation test	Erroneous data
Is the last character 'o' or 'e'?	ABC654321v

(b) Cerys will need an algorithm to search the table of stock data.

(i) State **two** advantages of using a linear search algorithm rather than a binary search algorithm.

(2)

1 .....

.....

2 .....

.....

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1. Question 5 · part c.ii · 2 marks

2020 NOV · Paper 02

Worked-example retry · Attempt part c.ii only.

(ii) State **two** disadvantages of using a linear search algorithm rather than a binary search algorithm.

(2)

1 .....

.....

2 .....

.....

(c) Cerys will use functions and procedures in her program.

(i) One similarity between a function and a procedure is that they are both subprograms.

State **two** other similarities.

(2)

1 .....

.....

2 .....

.....

(ii) Explain **one** difference between a function and a procedure.

(2)

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(Total for Question 5 = 12 marks)

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1. Question 5 · part c.ii · 2 marks · Working and answer

Working and answer

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2. Question 5 · part c.i · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part c.i only.

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5 Cerys Jones is the manager of a shop that sells electrical goods.

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(4)

Validation test	Erroneous data
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(i) State **two** advantages of using a linear search algorithm rather than a binary search algorithm.

(2)

1 .....

.....

2 .....

.....

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2. Question 5 · part c.i · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part c.i only.

(ii) State **two** disadvantages of using a linear search algorithm rather than a binary search algorithm.

(2)

1 .....

.....

2 .....

.....

(c) Cerys will use functions and procedures in her program.

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State **two** other similarities.

(2)

1 .....

.....

2 .....

.....

(ii) Explain **one** difference between a function and a procedure.

(2)

.....

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.....

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(Total for Question 5 = 12 marks)

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2. Question 5 · part c.i · 2 marks · Working and answer

Working and answer

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3. Question 3 · part d.iii · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.iii only.

3 Several encryption algorithms have been developed.

(a) Ahmed has started to create a Caesar cipher encryption program.

Caesar cipher encryption works by giving a number value to a key.

Each letter in a plaintext message is replaced by a new ciphertext letter using the key.

For example, as shown in the table, a key value of +2 would change the plaintext message **encrypt** to the ciphertext message **gpetarv**.

Plaintext	e	n	c	r	y	p	t
Ciphertext	g	p	e	t	a	r	v

Ahmed's program already converts from plaintext to ciphertext.

He wants to extend the program to:

- accept the input of a plaintext message
- accept the input of the key, which must be a positive integer between 1 and 25
- write the generated ciphertext to a text file named **Cipher.txt**
- display the ciphertext.

Open **Q03a** in the code editor.

Complete the program code.

You must use the structure given in the file **Q03a** to complete the program.

Do not add any further functionality.

Save your code as **Q03aFINISHED** with the correct file extension for the programming language.

(8)

(b) Explain **one** reason why the Caesar cipher is a weak encryption algorithm.

(2)

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3. Question 3 · part d.iii · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.iii only.

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(c) The Vigenere cipher encrypts plaintext using a series of Caesar ciphers, based on the letters of a keyword and a Vigenere square.

Figure 1 shows a Vigenere square.

	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
a	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
b	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a
c	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b
d	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c
e	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d
f	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e
g	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f
h	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g
i	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h
j	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i
k	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j
l	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k
m	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l
n	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m
o	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n
p	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
q	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p
r	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q
s	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r
t	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s
u	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t
v	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u
w	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v
x	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w
y	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x
z	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y

Figure 1

Complete the table to perform a Vigenere decryption of the ciphertext message **flmktrwhee** using the keyword **thirty**.

(4)

Encrypted letter	f	l	m	k	t	r	w	h	e	e
Keyword letter										
Decrypted letter										

P 6 6 4 9 0 A 0 9 1 6

9

Turn over ▶

3. Question 3 · part d.iii · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.iii only.

(d) A user-defined function uses a Vigenere cipher to encrypt plaintext into ciphertext.

This pseudocode shows how the function is called.

```
SET ciphertext TO ciphertext & getCipherLetter(keywordLetter, plaintextLetter)
```

(i) State what would be returned by this function.

(1)

.....

.....

(ii) Give the name of **one** parameter passed to this function.

(1)

.....

(iii) State what is meant by a **built-in subprogram**.

(1)

.....

.....

(Total for Question 3 = 17 marks)

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Working and answer

4. Question 3 · part b · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

3 Dates are often used in programs.

(a) A program accepts the number of a month and outputs the number of days in the month.

Open **Q03a** in the code editor.

Use the code to answer these questions.

(i) Identify the name of a local variable.

(1)

(ii) Identify the name of a data structure.

(1)

(iii) Identify the name of a built-in subprogram.

(1)

(iv) Identify the name of a user-written function.

(1)

(v) Identify the name of a parameter.

(1)

(vi) The program includes validation to ensure the month number is between 1 and 12 inclusive.

Complete the table by adding the type of test data that is being used to test the validation.

The first type of test has been completed for you.

(2)

Type of test data	Test data
Normal	3
	1
	15

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4. Question 3 · part b · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

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(b) Explain **one** reason why parameters are used to pass data into subprograms.

(2)

(c) A leap year is a year that has 366 days rather than 365 days. A leap year usually occurs every four years.

A year that is a leap year:

- can be exactly divided by 400

Or

- can be exactly divided by 4, but **not** exactly divided by 100.

These years are leap years: 2000, 2012 and 2020.

These years are not leap years: 1700, 1900 and 2021.

A program is needed that will accept the input of a year and output whether it is a leap year or not.

Open **Q03c** in the code editor.

Write the program.

You **must** use the structure given in **Q03c**.

Do not add any further functionality.

Save your code as **Q03cFINISHED** with the correct file extension for the programming language.

(6)

(Total for Question 3 = 15 marks)

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4. Question 3 · part b · 2 marks · Working and answer

Working and answer

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5. Question 2 · part a.i · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part a.i only.

2 Solutions to problems are made up of many different components.

(a) Programmers use subprograms when developing code.

(i) Give **two** reasons to use **subprogram libraries** when developing code.

(2)

1 .....

.....

2 .....

.....

(ii) Identify which **one** of these must return a result.

(1)

- ☒ **A** Function
- ☒ **B** Iteration
- ☒ **C** Procedure
- ☒ **D** Selection

(iii) State what is meant by the term **local variable**.

(1)

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5. Question 2 · part a.i · 2 marks · Working and answer

Working and answer

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6. Question 2 · part a.ii · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part a.ii only.

2 Solutions to problems are made up of many different components.

(a) Programmers use subprograms when developing code.

(i) Give **two** reasons to use **subprogram libraries** when developing code.

(2)

1 .....

.....

2 .....

.....

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(1)

- ☒ **A** Function
- ☒ **B** Iteration
- ☒ **C** Procedure
- ☒ **D** Selection

(iii) State what is meant by the term **local variable**.

(1)

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6. Question 2 · part a.ii · 1 mark · Working and answer

Working and answer

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Practice workbook · Local and global variables

Defining local and global variables, where each is created and where each can be accessed, and contrasting them.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T2-03

1. Question 1 · part e · 2 marks

2019 JUN · Paper 02

Worked-example retry · Attempt part e only.

1 Computer programs use several different programming constructs.

(a) Identify the **most appropriate** data type for the value 4.5

(1)

- ☒ **A** Boolean
- ☒ **B** Character
- ☒ **C** Real
- ☒ **D** String

(b) Give a definition of a logic error.

(1)

(c) Open **Q01c** in the code editor.

The program should print out a counter and the counter with 7 added to it.

There are **three** errors in the code.

Amend the code to correct the errors.

Save your amended code as **Q01cFINISHED** with the correct file extension for the programming language.

(3)

(d) Give **one** reason why computer programmers should use comments in their programs.

(1)

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1. Question 1 · part e · 2 marks

2019 JUN · Paper 02

Worked-example retry · Attempt part e only.

(e) Describe the difference between a local variable and a global variable in a computer program.

(2)

(f) Trevor is the manager of a shoe shop.

Sales assistants can earn a bonus based on the numbers of pairs of shoes they sell and the total income for the shop each day.

Open **Q01f** in the code editor.

Amend the code to complete the 'if statement' used to produce the outputs described in the table.

Condition	Output
Shop income is more than £5000 or sales assistant has sold at least 10 pairs of shoes	Bonus is 10% of salary
Shop income is £2000 or more and sales assistant has sold at least 5 pairs of shoes	Bonus is 5% of salary

Do not add any further functionality.

Save your amended code as **Q01fFINISHED** with the correct file extension for the programming language.

(4)

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P 6 1 8 8 1 R A 0 3 1 6

3

Turn over

1. Question 1 · part e · 2 marks · Working and answer

Working and answer

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2. Question 2 · part a.iii · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part a.iii only.

2 Solutions to problems are made up of many different components.

(a) Programmers use subprograms when developing code.

(i) Give **two** reasons to use **subprogram libraries** when developing code.

(2)

1 .....

.....

2 .....

.....

(ii) Identify which **one** of these must return a result.

(1)

- ☒ **A** Function
- ☒ **B** Iteration
- ☒ **C** Procedure
- ☒ **D** Selection

(iii) State what is meant by the term **local variable**.

(1)

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2. Question 2 · part a.iii · 1 mark

2023 JUN · Paper 02

Further practice · Attempt part a.iii only.

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(b) A school is planning a fish and chip dinner for the students and their families.

(i) Tickets are only sold to families.

A family is one to four adults and one to six children.

A program is being written to help manage ticket sales.

Complete the table to show examples of normal, boundary and erroneous numeric test data for the program.

(3)

	Adult	Children
Normal		
Boundary		
Erroneous		

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(ii) The kitchen staff use a program to determine whether they have enough chips or how many more they need to order.

Open **Q02bii** in the code editor.

There are **four** errors in the code.

Amend the code to correct the errors.

Use this test data to help you find the errors.

Chips in stock (kilograms)	Number of adults	Number of children	Expected output
19	100	150	Stocks available
15	100	150	Order 4.0 kilograms

Save your amended code as **Q02biiFINISHED** with the correct file extension for the programming language.

(4)

(Total for Question 2 = 11 marks)

2. Question 2 · part a.iii · 1 mark · Working and answer

Working and answer

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3. Question 4 · part a.i · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part a.i only.

4 Programmers write code that keeps track of values, is efficient and uses subprograms.

(a) Global variables are different from local variables.

(i) State where a **global** variable is created.

(1)

.....

.....

(ii) State where a **local** variable is accessible.

(1)

.....

.....

(b) An algorithm reports the range that an integer falls within.

Integers from 0 to 39 are in the low range.

Integers from 40 to 69 are in the middle range.

Integers from 70 to 100 are in the high range.

Figure 4 and **Figure 5** show two algorithms that solve this problem.

```
1  SEND 'Enter an integer: ' TO DISPLAY
2  RECEIVE theNumber FROM (INTEGER) KEYBOARD
3  IF (theNumber < 40) THEN
4      SEND 'Low range' TO DISPLAY
5  ELSE
6      IF (theNumber < 70) THEN
7          SEND 'Middle range' TO DISPLAY
8      ELSE
9          SEND 'High range' TO DISPLAY
10     END IF
11 END IF
```

Figure 4

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

8

P 7 5 7 3 7 A 0 8 1 6

3. Question 4 · part a.i · 1 mark · Working and answer

Working and answer

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4. Question 4 · part a.ii · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part a.ii only.

4 Programmers write code that keeps track of values, is efficient and uses subprograms.

(a) Global variables are different from local variables.

(i) State where a **global** variable is created.

(1)

.....

.....

(ii) State where a **local** variable is accessible.

(1)

.....

.....

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7          SEND 'Middle range' TO DISPLAY
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9          SEND 'High range' TO DISPLAY
10     END IF
11 END IF
```

Figure 4

DO NOT WRITE IN THIS AREA
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DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

4. Question 4 · part a.ii · 1 mark · Working and answer

Working and answer

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